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Tugas kalkulus

1.) $y = \cos^3 (x^2+1)$

$y = \cos^n (x^2+1) \rightarrow -n \cdot u' \cos^{n-1} u . \sin u$

$\rightarrow -3 \cdot 2x \cos^2 (x^2+1) . \sin (x^2+1)$

$\rightarrow -6x \cos^2 (x^2+1) . \sin (x^2+1)$

2.) $f(x) = (x^2+5x-8)^{75}$

$u = x^2+5x-8$

$y = u^{75}$

$u' = 2x+5$

$y' = 75u^{74}$

$y = f(u),\ u = g(v)$, dan $v = h(x)$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= (75u^{74})(2x+5)$

$= 75(x^2+5x-8)^{74}(2x+5)$

3. $f(t) = \sin(\cos t + 2t)^7$

$v = \cos(t) + 2t$

$u = \sin(v)$

$y = u^7$

$\frac{dy}{dt} = 7u^6 . \cos(v) . -\sin(t) + 2$

$= -7 . \sin^6 (\cos(t) + 2t) . \cos(\cos(t) + 2t) . \sin((t)) + 2t)$

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4. $y = (3x + 10)^{10}$

$u = 3x+10$

$y = u^{10}$

$u' = 3$

$y' = 10u^9$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{dy}{dx}$

$= (10u^9)(3)$

$= 10(3x+10)^9(3)$

5.) $z = 3x^2 - y^2$; $x = 2s + 7t$; $y = 5st$

$z' = 6x - 2y$

$x' = 2 + 7$

$y' = 5s$

$\frac{dz}{dt} = \frac{dz}{dx} \cdot \frac{dx}{dt} + \frac{dz}{dy} \cdot \frac{dy}{dt}$

$= (6x)(7) + (-2y).(5s)$

$= 42x + 105y$

$= 42(2s + 7t) - 108(55t)$

$= 84s + 294t - 505^2t$

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6.) $w = e^x \sin y + e^y \sin x \; ; \; x = 3t \; ; \; y = 2t$

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt} + \frac{dw}{dy} \cdot \frac{dy}{dt}$$

$$= (e^x \sin y + e^y \cos x)(3) + (e^x \cos y + e^y \sin x)(2)$$

$$= (e^{3t} \sin 2t + e^{2t} \cos 3t)(3) + (e^{3t} \cos 2t + e^{2t} \sin 3t)(2)$$

$$= 3e^{3t} \sin 2t + 3e^{2t} \cos 3t + 2e^{3t} \cos 2t + 2e^{2t} \sin 3t$$

$$= 3e^{3t} \sin 2t + 2e^{3t} \cos 2t + 3e^{2t} \cos 3t + 2e^{2t} \sin 3t$$

$$= e^{3t}(3 \sin 2t + 2 \cos 2t) + e^{2t}(3 \cos 3t + 2 \sin 3t)$$

7.) $w = x^2 y \; ; \; x = st \; ; \; y = s - t$

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt} + \frac{dw}{dy} \cdot \frac{dy}{dt}$$

$$= (2xy) \cdot (s) + (x^2)(-1)$$

$$= 2sxy - x^2$$

8.) $w = x^2 - y \ln x \; ; \; x = \frac{s}{t} \; ; \; y = s^2 t$

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt} + \frac{dw}{dy} \cdot \frac{dy}{dt}$$

$$= (2x - (y/x)) \cdot (-s/t^2) + (-\ln x)(s^2)$$

$$= \frac{-s(2x^2 - y)}{xt^2} - s^2 \ln x$$

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9) $z = x^2 y$; $x = 2t + s$; $y = 1 - st^2$

$$\frac{dz}{dt} = \frac{dz}{dx} \cdot \frac{dx}{dt} + \frac{dz}{dy} \cdot \frac{dy}{dt}$$

$$= (2xy)(2) + (x^2)(-2st)$$

$$= 4xy + (-2st x^2)$$

$$= 4xy - 2stx^2$$